DO NOW — RECALL

Lesson 100 · GPUs: Parallel Work Beyond Graphics

Name: _____ Class: _____ Date: _____

1. Which processor is designed for throughput on massive parallel workloads?

2. What is parallelism?

3. What is a data dependency?

4. An image array has 4 rows × 3 columns. How many pixels?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

GPU.

2. Model answer

Actually running independent work at same time.

3. Model answer

A task needs another task's result.

4. Model answer

12.

By the end, you can…

• GPUs provide many execution units designed for high-throughput parallel operations on large batches of data.

• Besides graphics, GPUs can accelerate machine learning, physical simulations, image processing and some scientific calculations.

• Transfers between memory/device, limited parallelism and divergent branching can outweigh GPU speedup on small or sequential tasks.

Vocabulary — know these terms

GPU — Graphics Processing Unit.

SIMT — Single instruction, multiple threads execution model.

throughput — Total work per time.

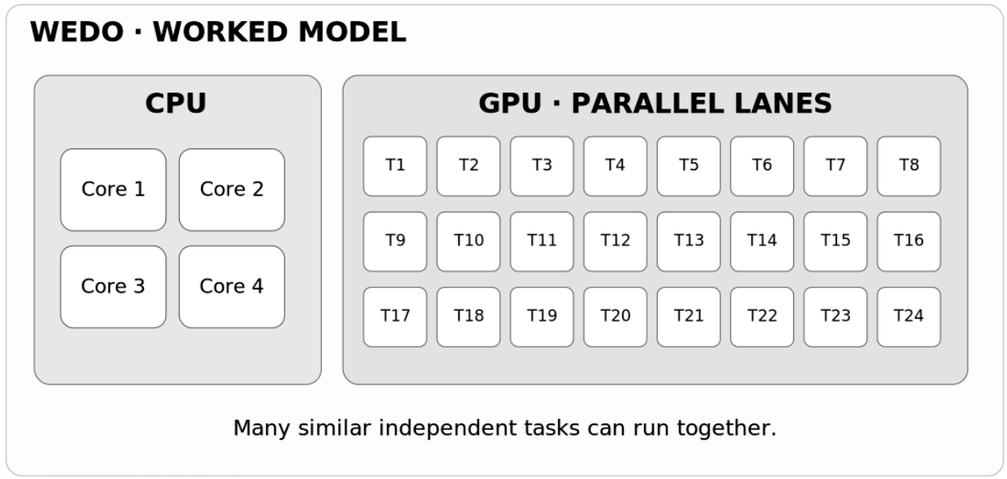
divergence — Threads follow differing control paths.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A sequential data-dependent algorithm calculates next item using previous result. Does GPU automatically accel

Not necessarily: dependence limits simultaneous work, and data-transfer overhead may dominate.



WEDO · WORKED MODEL; A sequential data-dependent algorithm calculates next item using previous result. Does GPU automatically accelerate it?; Not necessarily: dependence limits simultaneous work, and data-transfer overhead may dominate.

NEW LEARNING

Read, annotate and explain

Observe

GPUs provide many execution units designed for high-throughput parallel operations on large batches of data.

Apply

Besides graphics, GPUs can accelerate machine learning, physical simulations, image processing and some scientific calculations.

Evaluate

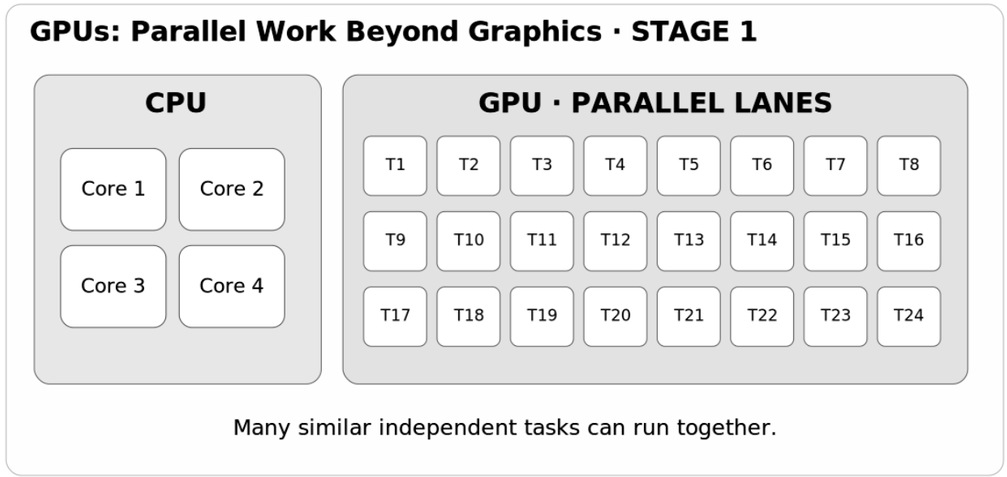
Transfers between memory/device, limited parallelism and divergent branching can outweigh GPU speedup on small or sequential tasks.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

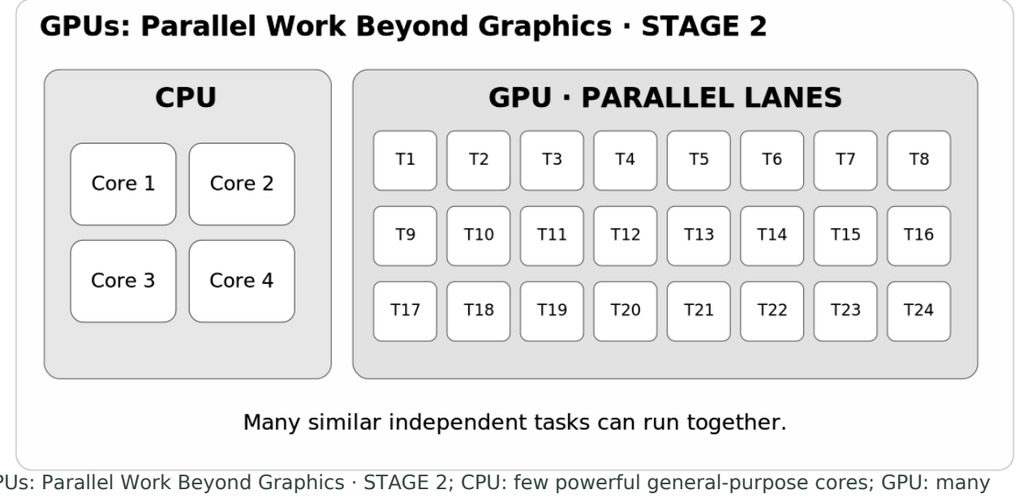
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



GPUs: Parallel Work Beyond Graphics · STAGE 1; CPU: few powerful general-purpose cores

New Learning



GPUs: Parallel Work Beyond Graphics · STAGE 2; CPU: few powerful general-purpose cores; GPU: many throughput-oriented parallel execution lanes

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Follow the reasoning
Not necessarily: dependence limits simultaneous work, and data-transfer overhead may dominate.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A GPU is used exclusively for drawing computer-game graphics.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

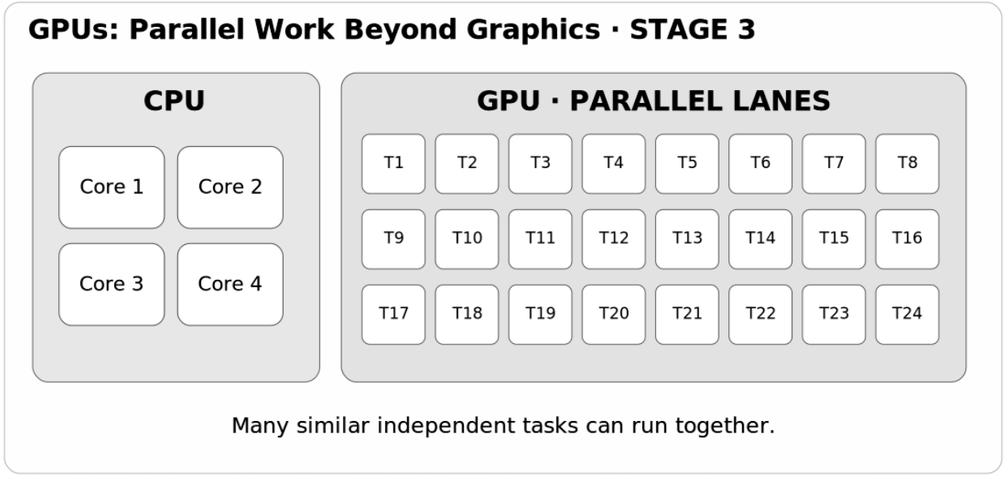
GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

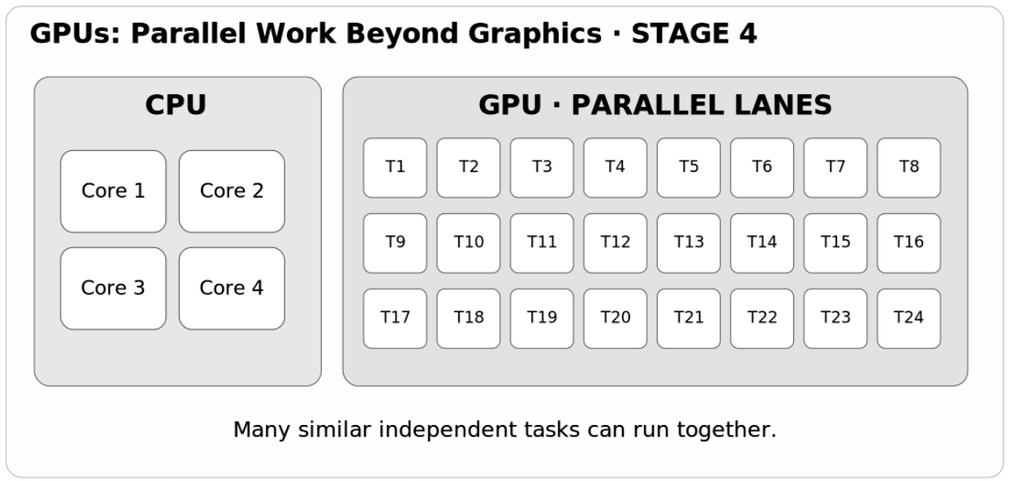


GPUs: Parallel Work Beyond Graphics · STAGE 3; CPU: few powerful general-purpose cores; GPU: many throughput-oriented parallel execution lanes; Image pixels can be processed independently

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



GPUs: Parallel Work Beyond Graphics · STAGE 4; CPU: few powerful general-purpose cores; GPU: many throughput-oriented parallel execution lanes; Image pixels can be processed independently; AI tensor work and scientific simulation also exploit GPUs

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare CPU and GPU for 12-item sequential task versus million-element independent matrix operations. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Explain two features that make a GPU useful for applying same filter to millions of pixels. [6]

Question 2

Give two non-graphics GPU applications and explain how parallelism helps. [6]

I DO — WORKED EXAMPLE

Follow the worked method

Explain why processing one million independent pixels may suit a GPU.

Follow the reasoning

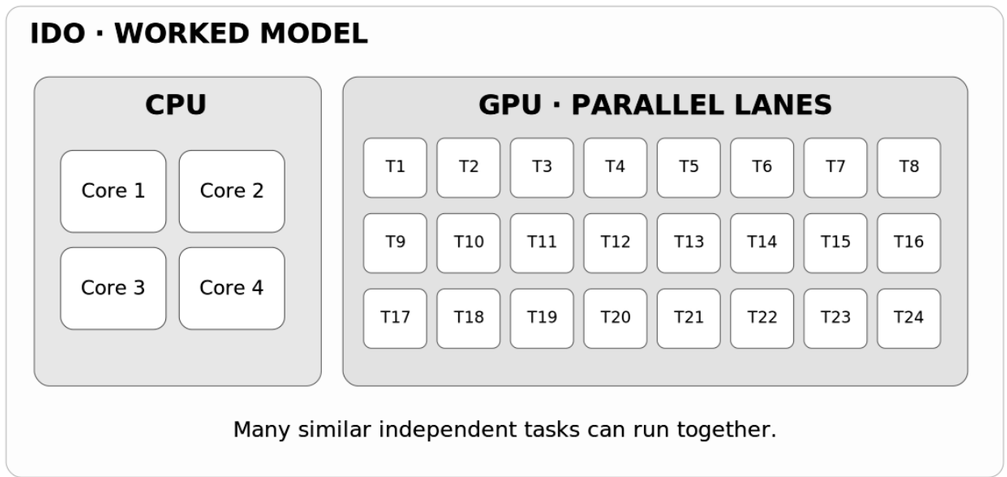
Each pixel can receive similar operation in parallel; many GPU threads process different positions.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Explain why processing one million independent pixels may suit a GPU.

Each pixel can receive similar operation in parallel; many GPU threads process different positions.



IDO · WORKED MODEL; Explain why processing one million independent pixels may suit a GPU.; Each pixel can receive similar operation in parallel; many GPU threads process different positions.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A sequential data-dependent algorithm calculates next item using previous result. Does GPU automatically accelerate it?